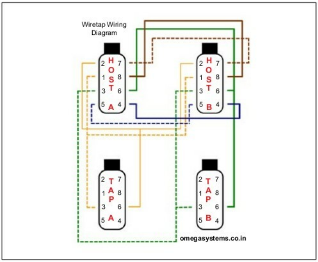

Figure 2: Wiretap

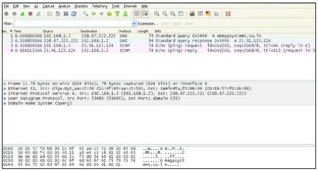

Figure 3: DNS query and ICMP echo request

3. *Using Switched Port ANalyzer (SPAN), also called port mirroring:* SPAN ports are available with some of the managed switches. Once enabled, this feature copies traffic from the defined ports to the SPAN port. The Wireshark system is connected to this SPAN port to capture the traffic. The disadvantage of this system is that the SPAN port can get overloaded since it receives all traffic from many ports. This results in loss of packets.
4. *Using ARP spoofing:* This requires using the ARP spoofing tool such as ettercap-NG, which is tricky and may result in network disruption if sufficient care is not taken.

The first capture

Let us start by capturing packets for the DNS (Domain Name Service) protocol. As you must be aware, DNS resolves URLs to IP addresses. It is a very simple UDP protocol (works on TCP as well). It sends out a DNS query and gets a DNS query response.

To capture DNS, start Wireshark and select the desired interface to start packet capture. Go to the command prompt under Windows (or the terminal, in the case of Linux) and ping any URL—the associated screenshot (Figure 3) is for *omegasystems.co.in*. You will see several captured packets on the screen. Stop packet capture.

Background information on Wireshark.org

In May 2006, Gerald Combs (the original author of Ethereal) went to work for CACE Technologies (best known for winpcap). Unfortunately, he had to leave the Ethereal trademarks behind. This left the project in an awkward position. The only reasonable way to ensure the continued success of the project was to change the name. This is how Wireshark was born.
Wireshark is a network protocol analyser. It lets you capture and interactively browse the traffic running on a computer network. It runs on most computing platforms including Windows, OS X, Linux and UNIX.

Wireshark's Sectools rating

Sectools.org, maintained by the Nmap Project, has been cataloguing the network security community's favourite tools for more than a decade.
It ranks Wireshark as the No 1 among network security tools and describes it as follows: "Wireshark (known as Ethereal until a trademark dispute in Summer 2006) is a fantastic open source multi-platform network protocol analyser. It allows you to examine data from a live network or from a capture file on disk. You can interactively browse the capture data, delving down into just the level of packet detail you need. Wireshark has several powerful features, including a rich display filter language and the ability to view the reconstructed stream of a TCP session. It also supports hundreds of protocols and media types."

The command line request to 'PING' *omegasystems.co.in* triggered a 78-byte UDP DNS query from the computer system 192.168.1.2 (IP address in your capture will be the same as the TCP/IP configuration of your capture system) towards DNS server 208.67.222.222. A 94-byte reply was received with the IP address of *omegasystems.co.in* as 23.91.123.124.

A 74-byte ICMP Echo Request is now sent to the resolved IP address of *omegasystems.co.in*, for which a 74-byte ICMP Echo reply is received.

Please view the *Time* column, and you will notice that approximately 0.3 seconds were required to receive replies for both these queries.

A word of caution

The test scenarios described in this series of articles are capable of revealing sensitive information such as login names and passwords. Some scenarios, such as using ARP spoofing, will disturb the network temporarily. Make sure to use these techniques only in a test environment or avail explicit written permission before using them in a live environment. END

By: Rajesh Deodhar

The author is an IS auditor and network security trainer-consultant for the last two decades. He is a BE in industrial electronics, as well as a CISA, CISSP, CCNA and DCL. He can be contacted at *rajesh@omegasystems.co.in*